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Peer-to-peer local electricity market platform pricing strategies for prosumers

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Abstract: Peer-to-peer (P2P) energy trading in the local electricity market (LEM) handles local energy trades among prosumers connected to the same distribution network. LEM is governed by prosumer's individual choices, self-management and neighbourhood interactions. This study investigates the design of P2P LEM platform based on sharing economy for non-professional prosumers and provides an exposition towards conceptual integration of LEM in the existing market. Design features of the P2P platform are based on two-sided matching market theory. Here, P2P platform provider, i.e. an aggregator, operates a two-sided market platform to match and set prices for prosumers having different willingness to buy/sell. The economic rationale of the P2P platform is highlighted by a variety of pricing strategies that the aggregator could offer, focusing on the benefits of aggregator and prosumers. As the total number of trades among peers is a crucial factor to determine aggregator profitability, pricing with a capability to induce prosumer surplus acts as an instrument for the aggregator to bind prosumers in P2P LEM.

Nomenclature

Sets and indices

M	set of prosumers, indexed by k
$B \subset M$	set of buyer peers, indexed by i
$S \subset M$	set of seller peers, indexed by j
ρ	market type, i.e. retail and LEM, $\rho = \{r, l\}$
θ	action type, i.e. import and export, $\theta = \{\text{im}, \text{ex}\}$

Parameters

U	utility for prosumer
A_k	budget of prosumer k
Q_k^{total}	total energy demand of prosumer k
Q_k^{gen}	self-energy generation by prosumer k
$\lambda_k^{\text{r},\theta}$	energy price for prosumer k while importing/exporting from/to retailer
$\bar{v}_i/\underline{v}_i$	upper/lower limit of valuation range provided by buyer peers to the aggregator
$\bar{v}_j/\underline{v}_j$	upper/lower limit of valuation range provided by seller peers to the aggregator
$v_i^{\text{actual}}/v_j^{\text{actual}}$	actual valuation by buyer/seller peers

Variables

$Q_k^{\rho,\theta}$	energy imported/exported by prosumer k from/to retail/LEM
$\lambda_k^{\text{l},\theta}$	energy price for prosumer k while importing/exporting from/to LEM
n	number of possible trades with a given set of buyer and seller
π_k	profit of prosumer k when it acts as a supplier
SP_i/SP_j	surplus of buyer/seller peer

1 Introduction

With the increasing deployment of renewable energy (RE) sources, in the form of rooftop solar photovoltaic, concentrating solar thermal and small wind turbine (400 W to 20 kW), energy consumer is evolving as a prosumer, i.e. consumer as well as producer of electricity [1]. Despite environmental benefits from intermittent RE, high penetration of prosumers in distribution network poses challenges for local grid balancing and energy balance of prosumers. Local electricity market (LEM) can reduce these imbalances, where prosumers and consumers (energy deficit prosumers) can trade energy within their neighbourhood, facilitating self and local demand–supply balance [2, 3]. For LEM, a wide range of possible market models, such as peer-to-peer (P2P), peer-to-grid and organised prosumer groups or community-based markets, are evolving [1]. These market models differ by the way they realise communication architecture, market, grid network and energy transfer. Among these, the P2P market model provides prosumer-centric benefits by facilitating direct participation/transactions among non-professional prosumers and providing them individual choices, self-management and platform for neighbourhood interactions [1, 4, 5].

P2P platforms can be designed based on (i) sharing economy, (ii) game theory or (iii) auction theory [5–7]. P2P platforms using sharing economy and online matching for off-grid prosumers are in the nascent stage of research and have been investigated through pilot projects like Piclo in the UK and Vandebron in the Netherlands [8, 9]. Business models of these projects vary in terms of business entities, customers needs, trading channels, value propositions and profit/revenue structure [9]. This restricts their direct applicability to grid-connected prosumers. Game theory and auction-based P2P energy market platforms can coordinate trading between grid-connected electric vehicle/distributed resources (DRs) and wholesale electricity markets [6, 7]. P2P trading based on game theory can motivate electric vehicle/DRs to cooperate, for common goals such as cost reduction, revenue maximisation and RE maximisation [4, 6, 7]. P2P platform based on auctions, such as discriminatory auction and double auction, can coordinate problems that arise from a large number of integrated DRs, and the traded price negotiations between an aggregator and prosumers [7, 9]. The bid/ask prices in auctions are based on game-theoretic models [7]. The centralised auctions and game theory-based platforms focus on competition or common goals rather than prosumer interaction. Prosumers interact and behave as per their non-professional attributes (like individual choices, self-management and neighbour interactions) as in P2P sharing economy, reflecting prosumer centricity [1, 10, 11]. However, P2P LEM based on sharing economy for grid-connected non-professional prosumers is yet to be explored, apart from

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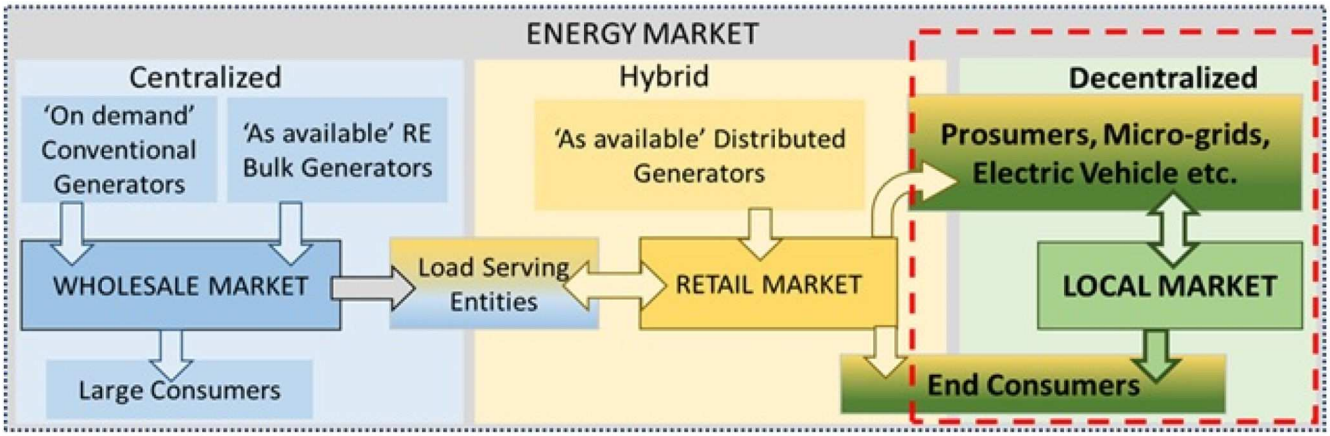

Fig. 1 *LEM integration and market organisation*

understanding their integration with existing centralised power systems.

In P2P transactions, each peer acts as a provider as well as a receiver of resources. P2P sharing is relevant for a range of sectors including online shopping, transport and lending, which promote small sellers to compete with prominent suppliers of goods and services [12]. The same can be applied to energy trading in prosumer-centric electricity markets, as prosumer upholds the characteristics of a peer. In a P2P trading market, a prosumer is connected to an online-platform via information and communication network. While searching for its appropriate peer match, prosumer takes significant time which adds to its search friction and related costs [12, 13]. Additionally, peers interact independently and make price decisions through bilateral negotiations at prices they are willing to buy/sell [3, 14]. As the utility function for every prosumer varies, its bargaining process would hinder price settlement, making markets inefficient [15, 16]. Thus, matchmaking (searching) and price negotiation between prosumers is a major concern in P2P sharing markets, which is the research focus of this paper.

Considering the literature, the research gap can be summarised as follows:

(i) The existing studies focus on P2P energy sharing based on auction and game-theoretic models. For non-professional players like prosumers, which are also grid connected, sharing economy based platform needs investigation. Besides, the conceptual integration of LEM in existing energy market needs to be explored.
(ii) The existing knowledge base is restricted to the concept and use of sharing economy based P2P platform (two-sided matching market) for general prosumers. Their implementation for electricity markets is yet to be modelled and experimented. Moreover, the design characteristics of the platform vary with its application. Application, implementation and benefit calculations for P2P LEM require wider investigations.

To handle the aforementioned research gaps, this paper proposes a two-sided sharing economy-based P2P matching for a LEM. In economics, a two-sided matching market is an intermediary or platform which enables interaction between two sets of agents [14]. In P2P LEM, the platform provider, i.e. an aggregator, facilitates value-creating interactions between external producer and consumers (prosumers) of electricity, by providing an open and participative infrastructure. Aggregator collects information from prosumers, performs search and match operations as well as suggests a common price for matched buyer–seller peer [15, 17]. Matched buyer–sellers would make final decisions (by accepting or rejecting) the price suggested by the aggregator. Aggregator charges prosumers for providing matching and pricing services. The choice and economy of aggregator's pricing decisions depend upon service type, location, value and decision characteristics of prosumers [15, 18]. This creates new challenges and motivation to investigate prosumer-centric platform for pricing strategies concerning P2P LEM. The key contributions of this paper are:

(i) Novel exposition of local energy trading through P2P LEM platform for grid-connected prosumers (based on sharing economy) and conceptual integration of developed LEM in existing electricity markets.
(ii) Assessment of aggregator's multiple pricing strategies for P2P energy trading in LEM and analysis of their impact on aggregator's profitability and prosumers surplus.

Rest of the paper is organised as follows: Section 2 discusses LEM organisation and P2P design. Prosumer modelling and trading framework are provided in Section 3. Two-sided platform pricing strategies are discussed in Section 4 while Section 5 presents the numerical example and economic discussion. Section 6 draws the conclusions. The Key symbols are indicated in the Nomenclature section.

2 LEM organisation and P2P design

This section provides a conceptual understanding of LEM integration in the existing energy market and sharing economy-based P2P.

2.1 LEM organisation in the existing energy market

LEM organisation is illustrated in Fig. 1 by considering the existing wholesale and retail market, as a whole. This combination synergises benefits in response to the present and future needs of the energy market.

Management of energy markets is carried out in centralised, hybrid and decentralised manner. In the centralised wholesale market, 'as available' RE plants and 'on demand' dispatchable bulk power plants (collectively called energy sellers) sell electricity to bulk energy buyers like load-serving entities. In the hybrid retail market (a hybrid of centralised and decentralised market), load-serving entities interact with the wholesale market, other retailers and 'as available' RE-based distributed generation, to provide electricity to consumers [19]. In decentralised LEM, local energy sources such as prosumers, micro-grid or electric vehicle actively trade their energy according to their demand, generation, value, priority, available budget etc. and make self-decisions. End consumers can take power from the retail or local market to fulfil their energy requirement, as shown in Fig. 1. Moreover, wholesale markets deal with business-to-business (B2B) transactions, retail markets address business-to-consumer and B2B relations and LEM have consumer-to-consumer (C2C) transactions.

There are various ways to link business models of LEM with existing markets. One way is to link with the retail supplier platform where existing suppliers offer a LEM trading platform and operate as a local community [11]. Alternatively, energy contracts are offered between energy generators (or battery storage), energy retailers/suppliers acting as intermediatories and prosumers to trade electricity [20]. However, the challenge is retailers/suppliers are professional entities, whereas prosumers are a non-professional entity. A transaction can only be qualified as

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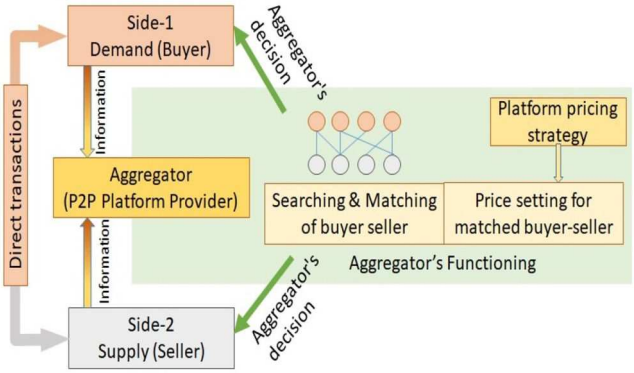

Fig. 2 *P2P LEM operation and aggregator's functioning*

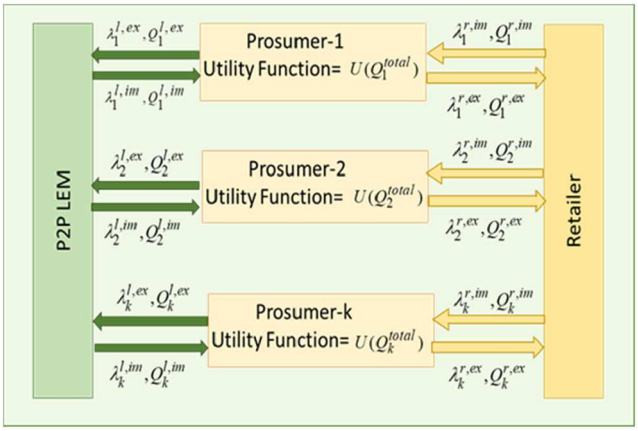

Fig. 3 *Prosumers with retail and P2P LEM*

C2C or P2P when it is undertaken by two non-professional entities [5].

Therefore, this paper considers, the prosumer transaction in LEM as P2P energy trading, undertaken by non-professional entities and focusing on sharing economy. Prosumers switch their roles between supplier and consumer during local trading. Supplier prosumer commits a certain quantum of energy for the distribution network and buyer prosumer commits to extract energy from the network. The injection and extraction of electricity between two prosumers need to be matched and designed using the two-sided matching market. During energy trading, matched prosumer pair signs delivery time agreement with each other. After trading in LEM, remaining untraded (excess/deficit) quantum is provided by retailer offering various contracts to prosumers. The exact timeline of the whole trading system depends on communication devices, the response time of connected energy resources and market/network policies which evolve over time. These issues are beyond the scope of this study.

2.2 P2P LEM design using sharing economy

Decentralised LEM operations require an interactive platform to express energy mismatch and facilitate P2P energy transactions among prosumers. In the P2P model of LEM, all peers (prosumers) communicate with a central node, known as aggregator [7, 14]. Aggregators in P2P commodity and service markets such as UBER, Airbnb and Grofers are commercial aggregators. They provide an online platform to collect, combine, assemble data or information from various sources and provide an interface to access this information [7]. Inspired by these market mechanisms, aggregator in P2P LEM would collect information from connected prosumers for electricity exchange/transaction, as shown in Fig. 2. Key actors in P2P LEM are

- *Seller peer*: Set of prosumers acting as energy providers.
- *Buyer peer*: Set of prosumers acting as consumers.
- *Aggregator*: It owns and manages the two-sides of the trading platform, i.e. buyer and seller side, as shown in Fig. 2. It acts as an intermediary between prosumers and performs three main functions, namely searching, matching and pricing [7].

Based on peer information, aggregator operates P2P LEM as a two-sided matching market to search and match prosumer peers [7, 15]. As per platform pricing strategy, aggregator calculates and communicates prices for various matched prosumer pairs. Aggregator acts as a profit-making entity and charges prosumers for providing the platform and related services.

In LEM, prosumer calculates the quantum of energy mismatch, time-duration and its corresponding valuation over time (referred as characteristics). It submits bids and offers in terms of quantum of energy to buy or sell, location and valuation to the platform. The aggregator then matches prosumer based on submitted bids/offers and provides pricing information to matched prosumers as per platform pricing strategy. Matched prosumer pair then accept/reject the price offer. If rejected, prosumer needs to connect to the platform again with revised valuation or characteristics.

3 Prosumer modelling and interface with the trading platform

This section models prosumers characteristics (both as consumer or supplier) and interaction with the trading platform.

3.1 Prosumer characteristic modelling

Prosumer monitors, control and optimises its households' assets (generation and load) using a home energy management system. Due to the RE generation and load uncertainty, prosumer faces energy imbalances (deficit or excess). During energy deficit, self-generation is less than its demand and prosumer acts as a consumer to import energy either from the retailer and/or from LEM, as shown in Fig. 3. During energy excess from its generation, prosumer acts as a supplier and exports energy to LEM.

Prosumer may own various RE technologies, such as wind and photovoltaic. Based on wind speed and solar irradiation data, prosumer calculates equivalent wind and PV output power generation. Total generation capacity of prosumers k is assumed to be G_k (kW) and Q_k^{gen} denotes the maximum available RE for prosumer k in a particular time. Prosumer behaviour as consumer and supplier can be modelled mathematically as mentioned in the following sections

3.1.1 Prosumer as a consumer: Prosumer after utilising its self-generation, fulfils its remaining energy demand by purchasing energy through retailer and/or LEM, to maximise its utility function as

$$\max\ U(Q_k^{\text{total}}) \tag{1}$$

$$\text{s.t.}\quad \lambda_k^{\text{r,im}} Q_k^{\text{r,im}} + \lambda_k^{\text{l,im}} Q_k^{\text{l,im}} \le A_k \tag{2}$$

$$Q_k^{\text{r,im}} + Q_k^{\text{l,im}} + Q_k^{\text{gen}} = Q_k^{\text{total}} \tag{3}$$

At any instant, prosumer k buys energy $Q_k^{\text{r,im}}$ from the retailer at price $\lambda_k^{\text{r,im}}$ and/or energy $Q_k^{\text{l,im}}$ at price $\lambda_k^{\text{l,im}}$ from LEM. Summation of price–quantity product of all purchases reflect the prosumer cost, which should be less than its defined budget A, as shown in (2). Prosumer's total demand Q_k^{total} is satisfied from retailer supply $Q_k^{\text{r,im}}$, LEM supply $Q_k^{\text{l,im}}$ and self-generation supply Q_k^{gen} as per (3). Cost of self-generation is not considered as prosumer intends to install RE for own use only.

3.1.2 Prosumer as a supplier: After utilising its self-generation, a prosumer sells its excess energy generation in LEM to earn profit as

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$$\max \quad \pi_k = \lambda_k^{\mathrm{l,ex}} Q_k^{\mathrm{l,ex}} + \lambda_k^{\mathrm{r,ex}} Q_k^{\mathrm{r,ex}} \tag{4}$$

$$\text{s.t.} \quad Q_k^{\mathrm{l,ex}} + Q_k^{\mathrm{r,ex}} \le Q_k^{\mathrm{gen}} - Q_k^{\mathrm{total}} \tag{5}$$

here $\lambda_k^{\mathrm{l,ex}}$ and $\lambda_k^{\mathrm{r,ex}}$ are prices of selling $Q_k^{\mathrm{l,ex}}$ and $Q_k^{\mathrm{r,ex}}$ are the energies in LEM and retail, respectively. The quantity traded in LEM and retail should be less than or equal to the difference of generation and its demand.

As prosumer is a combination of consumer and supplier, the two models need to be combined to develop prosumer modelling. Consumer utility in (1) is an ordinal number and generator profit in (4) is a cardinal number. The two objectives cannot be directly added for modelling prosumers. Therefore, to represent consumer model in similar units as to those of supplier model, (1) is changed to its dual model as

$$\min \ \lambda_k^{\mathrm{r,im}} Q_k^{\mathrm{r,im}} + \lambda_k^{\mathrm{l,im}} Q_k^{\mathrm{l,im}} \tag{6}$$

$$\text{s.t.} \quad U(Q_k^{\mathrm{total}}) \ge U_k \tag{7}$$

$$Q_k^{\mathrm{r,im}} + Q_k^{\mathrm{l,im}} + Q_k^{\mathrm{gen}} = Q_k^{\mathrm{total}} \tag{8}$$

The dual utility maximisation is now expenditure (cost) minimisation problem as a function of prices, as shown in (6). The utility function of prosumers over its total demand $U(Q_k^{\mathrm{total}})$ will be greater than or equal to U_k, i.e. the minimum utility that a prosumer wants to achieve as shown in (7). The maximisation problem of prosumer as a supplier in (4) is changed to minimisation problem as

$$\min \ -\pi_k = -\lambda_k^{\mathrm{l,ex}} Q_k^{\mathrm{l,ex}} - \lambda_k^{\mathrm{r,ex}} Q_k^{\mathrm{r,ex}} \tag{9}$$

To model characteristics as a prosumer, (6) and (9) can be combined as

$$\min \ \lambda_k^{\mathrm{r,im}} Q_k^{\mathrm{r,im}} + \lambda_k^{\mathrm{l,im}} Q_k^{\mathrm{l,im}} - \lambda_k^{\mathrm{l,ex}} Q_k^{\mathrm{l,ex}} - \lambda_k^{\mathrm{r,ex}} Q_k^{\mathrm{r,ex}} \tag{10}$$

s.t. (5), (7) and (8)

In the first two terms of (10), prosumer acts as a consumer and minimises its energy procurement cost. Prosumer's willingness-to-pay (W2P), i.e. the maximum money a prosumer (as a consumer) is willing to pay for particular quantity, at a given time and budget, would influence its purchasing decision [21]. W2P decision for prosumer k is characterised by utility function and is modelled as

$$W2P_k = \lambda_k^{\mathrm{r,im}} Q_k^{\mathrm{r,im}} + \lambda_k^{\mathrm{l,im}} Q_k^{\mathrm{l,im}} \le U_k \tag{11}$$

Prosumer has some valuation to identify the optimal value it perceives for energy as per its energy function. This helps it to identify its utility and trade in LEM accordingly. In context to LEM, W2P for $Q_k^{\mathrm{l,im}}$ should be equal to (or less than) the price paid, i.e. $\mathrm{W2P}_k \le \lambda_k^{\mathrm{l,im}} \cdot Q_k^{\mathrm{l,im}}$.

The third and fourth term of (10) define prosumer energy exported to LEM and retail at a particular time instant while acting as a supplier. Export decisions in LEM are influenced by its valuation (in terms of willingness-to-earn (W2E)). The price $\lambda_k^{\mathrm{l,ex}}$ would reflect the revenue for $Q_k^{\mathrm{l,ex}}$ energy. Therefore, W2E should be equal to (or less) than the revenue obtained in LEM, i.e. $\mathrm{W2E}_k \ge \lambda_k^{\mathrm{l,ex}} . Q_k^{\mathrm{l,ex}}$.

3.2 Prosumer's interaction with aggregator in LEM

According to $\mathrm{W2P}_k$ and $\mathrm{W2E}_k$, prosumer interacts with aggregator through the online trading platform. Let M, $B \in M$ and $S \in M$ denote the set of prosumers, buyers and sellers, respectively. Each buyer $i \in B \subset M$ has an actual valuation $v_i^{\mathrm{actual}} \ge 0$ per unit of electricity (actual valuation refers to an estimate of its worth, i.e. W2P) and $v_i^{\mathrm{actual}} \le \lambda_k^{\mathrm{l,im}}$ [22]. Similarly, each sellee $j \in S \subset M$ has the actual valuation $v_j^{\mathrm{actual}} \ge 0$ and $v_j^{\mathrm{actual}} \ge \lambda_k^{\mathrm{l,ex}}$ for selling electricity. Each prosumer is assumed to trade one unit of electricity, i.e. $Q_i^{\mathrm{l,im}} = Q_j^{\mathrm{l,ex}} = 1$. Prosumer shares this information to the platform.

Actual valuation is buyer/seller personal information, to this exact valuation, aggregator specifies a range of prices (lower and upper limit) to buyer and seller peers. Buyer and seller peer select the valuation range as per their evaluation towards $\mathrm{W2P}_k$ and $\mathrm{W2E}_k$. $\underline{v}_i$ and $\bar{v}_i$ denote lower and upper valuation range for buyers, known to the aggregator. Similarly, the lower and upper valuation range of sellers is denoted by $\underline{v}_j$ and $\bar{v}_j$. The upper limit of valuation range of buyer $\bar{v}_i$ should be equal to v_i^{actual}. Similarly lower limit of valuation range of seller $\underline{v}_j$ should be equal to v_j^{actual}. Within the valuation range, aggregator search and match prosumers with similar interest.

$$v_i \ge v_j \tag{12}$$

Algorithm 1 (see Fig. 4) details the search and match algorithm of the aggregator. This algorithm identifies the buyer peer to be linked with a specific seller peer for feasible trade, as per available valuation. A matching H of the bipartite graph is a set of edge such that no two edges share their endpoints. In other words, the graph's vertices are buyers and sellers, and there is an edge (possible connection) between them if their characteristics are the same. As one seller/buyer peer can be linked with several buyer/seller peers, matching algorithm is designed using bipartite graph theory, i.e. a set of sellers/buyers is chosen such that two sellers/buyers are not paired with one buyer/seller [23, 24]. Mathematically, a matching

Input:	Buyer $B(Q_i, \underline{v}_i, \overline{v}_i)$ and seller $S(Q_j, \underline{v}_j, \overline{v}_j)$ set information *i.e.* quantity and valuation range as bipartite graph.
Output:	Matched buyer-seller pairs H
1:	Count number of buyer and seller in the graph.
2:	**If** $(B \ge S)$ Form a matrix X with row of buyer and column of seller satisfying Eq. (12) with 0 and 1 and edge $(B,S) \in E = 1$, indicating a possible match in the row of buyer and column of seller. If $(B < S)$ do opposite to form matrix X.
3:	$H := \phi$
4:	**While size** (S)
5:	For maximal matching- Calculate sum of X column wise giving preference to lower count
6:	**If** sum of $X \sim 0$
7:	Find the column with minimum value of sum other than 0
8:	For that column, look for row position having 1 to that column and give preference to first element of row (first come first serve priority) as a buyer.
9:	Allot (B, S) in H and assign 0 to row B and column S
10:	**Else if**
11:	Break
12:	**End while**

Fig. 4 *Algorithm 1: Search and match algorithm*

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Input: Buyer $B(Q_i, \overline{v}_i, \underline{v}_i)$, Seller $S(Q_j, \overline{v}_j, \underline{v}_j)$ and quantification of past characteristics μ_i, μ_j of buyer and sellers.

Output: p_j^{cp}, p_i^{cp}

1: **For each** matched buyer i and seller j

2: Initialize μ_i, μ_j and calculate transaction price for buyer $p_i^{cp} = \mu_i + (\overline{v}_i - \underline{v}_j)/2$

3: Initialize μ_j and calculate transaction price for seller $p_j^{cp} = \mu_j + (\overline{v}_i - \underline{v}_j)/2$

4: Check $p_i^{cp} \le p_j^{cp}$, if yes then update μ_i, μ_j and go to Step 2.

5: **End for**

Fig. 5 *Algorithm 2: Price setting algorithm*

H of the graph is a subset of the edges E, so that there is no vertex incident on more than one edge in H. This means that no two edges have a shared vertex. If all the vertices of the graph are covered by a matching H, then that H is a perfect matching. The output of the matching algorithm is a set of prosumer pair which will share energy.

Aggregator chooses the seller and buyer on a first come first serve basis. Thus, the seller/buyer who comes first would get priority. Aggregator informs the linked buyer–seller pair with a specific price, calculated using valuation range of matched prosumers and various pricing strategies, as discussed in Section 4. Buyer–seller pair can accept or reject aggregator's offer according to the price offered by the aggregator. In this paper, one buyer is linked with one seller as trading quantity is assumed to be fixed, i.e. $Q_i^{\text{l.im}} = Q_j^{\text{l.ex}} = 1$. One buyer can be matched with two or more seller to utilise full quantity locally, however, it is out of the scope of this paper. Moreover, aggregator focuses only on virtual energy trading and assumes that physical energy network, i.e. distribution grid is available for P2P local energy transactions.

4 Two-sided platform pricing strategies

The critical problem for an aggregator in a two-sided market is to design pricing for both sides to satisfy prosumers [15]. Choice of pricing strategy for the aggregator is mainly affected by two factors, namely (i) pricing objectives of platform and (ii) pricing strategy determinants [18]. Pricing objectives could be to maximise or maintain profit, revenue, market share etc. Determinants of pricing strategy could be customer surplus (monetary benefit) or customer price sensitivity [18]. In this work, the pricing strategy intends to maximise aggregator profit. Customer surplus is considered as a determinant of pricing strategy.

4.1 Pricing objective of platform: aggregator profit

Prosumers are characterised by an individual utility function, and their energy buying/selling price depends upon time (real-time or minutes ahead), technology, comfort etc. This requires segmentation of prosumers into groups having similar visualisation towards W2P/W2E. Aggregator sets a price according to valuation. Cost of online-platform is assumed to be negligible for the aggregator. Therefore, it has only the revenue side in its profit equation. Aggregator's profit equation under various price structures in a two-sided market platform is discussed.

4.1.1 Commission-based system: A commission is a service charge given by prosumer to the aggregator for providing platform and pricing services. Commission can be charged in the following three ways:

- *Percentage basis:* In percentage commission, aggregator receives a percentage of the transaction price for each successful trade [25]. The aggregator's profit π_{p} on a percentage basis is

$$\pi_{\text{p}} = \sum\nolimits_n rP_n \tag{13}$$

where r is commission rate and P_n is transaction price for pairing n number of buyer–seller pairs. The transaction price for each matched pair is calculated using the mid-market rate (MMR) [26] as

$$P_n = (\bar{v}_i - \underline{v}_j)/2 \tag{14}$$

After subtracting aggregator's commission, the clearing price for seller and buyer is denoted as p_j^{cp} and p_i^{cp}, respectively. Aggregator maximises the number of buyer–seller pairs and prefers matching of buyer–seller pair of high valuation range, to maximise profit, with fixed commission rate.

- *Flat fee:* In flat fee, aggregator receives a fixed amount as a fee for successful trade [25]. The profit π_{f} is

$$\pi_{\text{f}} = n \,.\, P_{\text{f}} \tag{15}$$

where P_{f} is the fixed fee per trade and n is the total number of trades. The transaction price for each matched pair is calculated using MMR as (14). After subtracting aggregator's profit, clearing price for buyer and seller peers are denoted as p_j^{cp} and p_i^{cp}, respectively. Aggregator maximises the number of buyer–seller pairs, to increase profit. This pricing strategy provides transparency as the buyer and seller are aware of the aggregator fee in advance.

- *Net listing:* Aggregator maximises profit by setting clearing price generally equal to the actual valuation of matched buyer–seller pair. Therefore, clearing price is decided as per th upper limit of valuation range $\bar{v}_i$ of buyer and lower limit of valuation range $\underline{v}_j$ of seller. The clearing price for buyer and seller are denoted as $p_j^{\text{cp}} = \underline{v}_j$ and $p_i^{\text{cp}} = \bar{v}_i$, respectively. Aggregator's profit π_{nl} from matching seller j to buyer i is

$$\pi_{nl} = \sum_n \bar{v}_i - \underline{v}_j \tag{16}$$

4.1.2 Discriminatory pricing: In discriminatory pricing, aggregator sets a unique commission rate for each matched buyer–seller pair [18, 26]. It prefers each buyer–seller according to their past characteristics related to transactions prices and prosumer behaviour. In addition, the aggregator can adjust prices according to demand and supply situation (i.e. number of buyer–seller). This reduces demand–supply gap and enhances the efficiency of the matching system.

The aggregator has its commission strategy acting as a multiplication weight (denoted by μ_i, μ_j) for each buyer–seller to conduct a trade. Ten clearing prices p_j^{cp} and p_i^{cp} are set for seller and buyer accordingly. Prices are determined such that buyers and sellers accept the price, to make trade successful. Algorithm 2 (see Fig. 5) provides an overview of price-setting algorithm. As previous data of prosumer is not known, therefore μ_i, μ_j for buyer and sellers is considered randomly. Aggregator profit π_d after successful trade between n buyer–seller pair is

$$\pi_d = \sum\nolimits_n p_j^{\text{cp}} - p_i^{\text{cp}} \tag{17}$$

4.2 Pricing strategy determinant: prosumer surplus

To model prosumer centricity, prosumer satisfaction is important to motivate prosumer to trade in P2P LEM. Prosumer surplus provides an economic scale to determine benefit received by a prosumer as a buyer/seller from a trade. Seller surplus SP_j is the

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Table 1 Prosumer data

Buyer peer index	$\underline{v}_i/\bar{v}_i$, INR	v_i^{actual}, INR	Seller peer index	$\underline{v}_j/\bar{v}_j$, INR	v_j^{actual}, INR
1	2–3	3	1	1–2	1
2	2–3	3	2	4–5	4
3	4–5	5	3	2–3	2
4	1–2	2	4	3–4	3
5	3–4	4	5	2–3	2

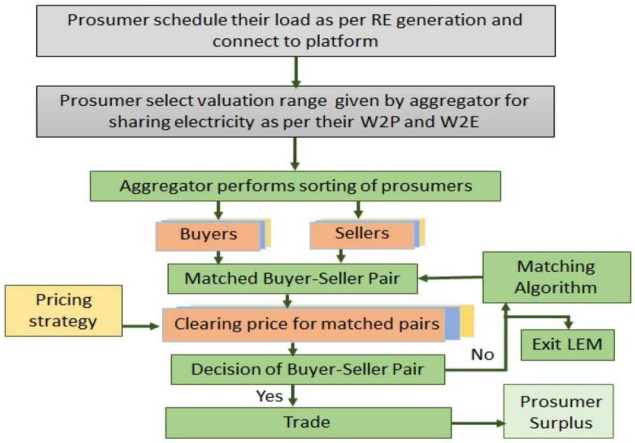

Fig. 6 *Trading process in P2P LEM*

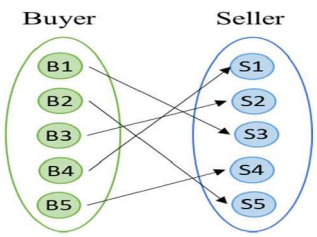

Fig. 7 *Matched results*

difference between the value it received (revenue) p_j^{cp} and the actual valuation for one unit of electricity at a given time. Similarly, buyer surplus SP_i is difference between price paid p_i^{cp} and its actual valuation. Seller and buyer surplus from trade are given by (18) and (19), respectively

$$\text{SP}_j = p_j^{\text{cp}} - v_j^{\text{actual}} \tag{18}$$

$$\text{SP}_i = v_i^{\text{actual}} - p_i^{\text{cp}} \tag{19}$$

Total surplus of buyer and seller TS from trade is

$$\text{TS} = \text{SP}_j + \text{SP}_i \tag{20}$$

P2P LEM platform designs promote the prosumers to trade and share energy locally. For this, the platform pricing strategy should increase aggregator's profits as well as prosumer surplus.

In addition to (18–20), prosumer's satisfaction can also be measured with market performance indices [25, 27]. Therefore, seller satisfaction index (SSI) and buyer satisfaction index (BSI) can be defined as in (21) and (22), respectively.

$$\text{SSI}_j = \frac{\sum_{i=1}^{\text{S}} Q_j^{\text{l,ex}} \cdot p_j^{\text{cp}}}{Q_j^{\text{l,ex}} \cdot v_j^{\text{actual}}} \tag{21}$$

$$\text{BSI}_j = \frac{Q_j^{\text{l,ex}} \cdot v_j^{\text{actual}}}{\sum_{j=1}^{B} Q_j^{\text{l,ex}} \cdot p_j^{\text{cp}}} \tag{22}$$

A higher value of SSI and BSI indicates higher satisfaction of sellers and buyers, respectively.

5 Numerical example and discussion

The platform's problem, in general, is to search for an appropriate peer match and an optimal price considering the prosumer valuation. Ten prosumers (five buyers and five sellers) are considered. Prosumers calculate their W2P and W2E, as per their characteristics using (10). It has actual valuation v_i^{actual} and v_j^{actual} as per their characteristics which is private information. The aggregator provides a valuation range $\underline{v}_j/\bar{v}_i$ and $\underline{v}_j/\bar{v}_j$ to trade and capture prosumer characteristics in LEM, which is randomly selected. From valuation range, prosumer provides approximation regarding its W2P and W2E. This information has been selected randomly and given in Table 1.

Search and match model is programmed in MATLAB® using Algorithm 1 (Fig. 4). The sequence and complete trading process are given in Fig. 6. The matched buyer–seller peers from search and match model are shown in Fig. 7. With matched buyer–seller pair, aggregator's profit and buyer–seller surplus is calculated by modelling pricing strategies in MATLAB®. While designing, algorithms have been evaluated on various data sets to confirm their feasibility and correctness for all conditions.

5.1 Percentage basis

In (13), r is fixed at 20% (10% from each side). Fig. 8*a* provides the clearing price for buyer and its corresponding matched seller. The transaction price is considered as an average valuation range, i.e. MMR and clearing prices are calculated by deducting aggregator's profit. Buyer and seller surplus is calculated using (18) and (19). Considering the match with the highest valuations, aggregator maximises profit by matching buyer 3 with seller 2, as shown in Fig. 9. This reduces the surplus of matched buyer–seller pair. Therefore, as clearing price increases, aggregator profit increases and buyer–seller surplus decreases.

5.2 Flat fee

Flat fee, P_{f} is assumed to be 0.25 per trade in (15). This is charged in total from buyer and seller, i.e. 0.125 from each. Therefore, th aggregator's profit per trade is fixed. Clearing price is the average valuation range and shown in Fig. 8. Buyer and seller surplus is higher as compared to the fixed aggregator's profit, as shown in Fig. 9. This pricing strategy is independent of clearing price and therefore, the profit of buyer–seller increases.

5.3 Net listing

In net listing, different clearance price is obtained for buyer and seller, shown in Fig. 9. Aggregator earns a profit by minimising buyer–seller surplus. Therefore, clearing price set for the buyer is in the upper limit of valuation range and for the seller it is in the lower limit of the valuation range. This strategy enhances benefits to the aggregator and offers no surplus to buyer and seller, as shown in Figs. 8 and 9.

5.4 Discriminatory pricing

In discriminatory pricing, aggregator sets a different price for paired buyer–seller, as shown in Fig. 8*d*. The transaction price is set with price-setting Algorithm 2 (Fig. 5). Buyer–seller surplus is

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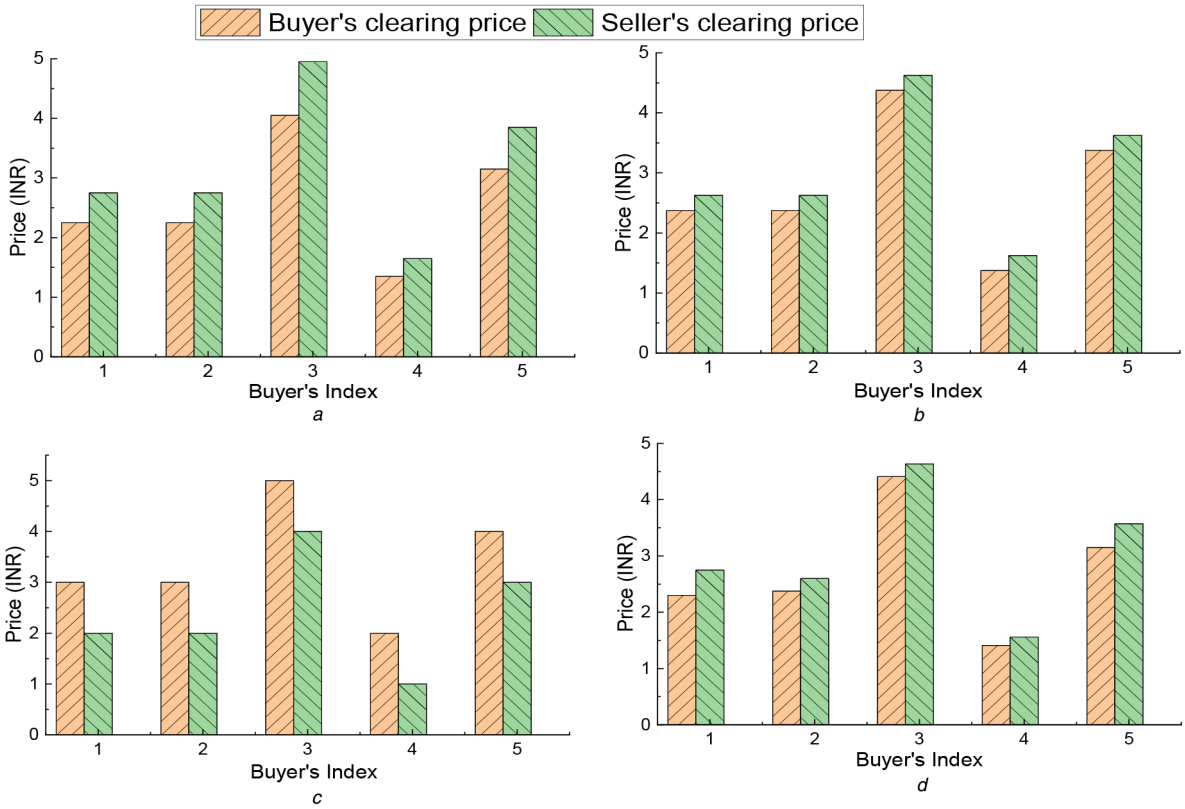

Fig. 8 *Market clearing price under different pricing strategies*
(a) Percentage basis, ***(b)*** Flat fee, ***(c)*** Net listing, ***(d)*** Discriminatory pricing

Fig. 9 *Surplus of buyer–seller and aggregator's profit under various pricing strategies*
(a) Percentage basis, ***(b)*** Flat fee, ***(c)*** Net listing, ***(d)*** Discriminatory pricing

not equally divided and depends on market conditions such as the number and valuation of buyer–seller. The commission strategy μ is considered randomly which is [0.08, 0.05, 0.02, 0.06, 0.1] for buyers and [0.1, 0.04, 0.03, 0.04, 0.02] for corresponding matched seller. Aggregator's profit is unique from each buyer–seller and slightly more than the total buyer–seller surplus, as per Fig. 9.

5.5 Comparison between different pricing strategies

Each strategy designs the clearing price differently. This case study highlights that net listing strategy provides maximum benefit to the aggregator, as shown in Figs. 9 and 10. This strategy neglects buyer and seller surplus. BSI and SSI are 1 as shown in Table 2. If buyer–seller surplus is zero, the buyer would be uninterested between choosing LEM or retail market and similarly, the seller would be unconcerned about sharing energy. Flat fee and percentage-based pricing strategy consider buyer–seller surplus,

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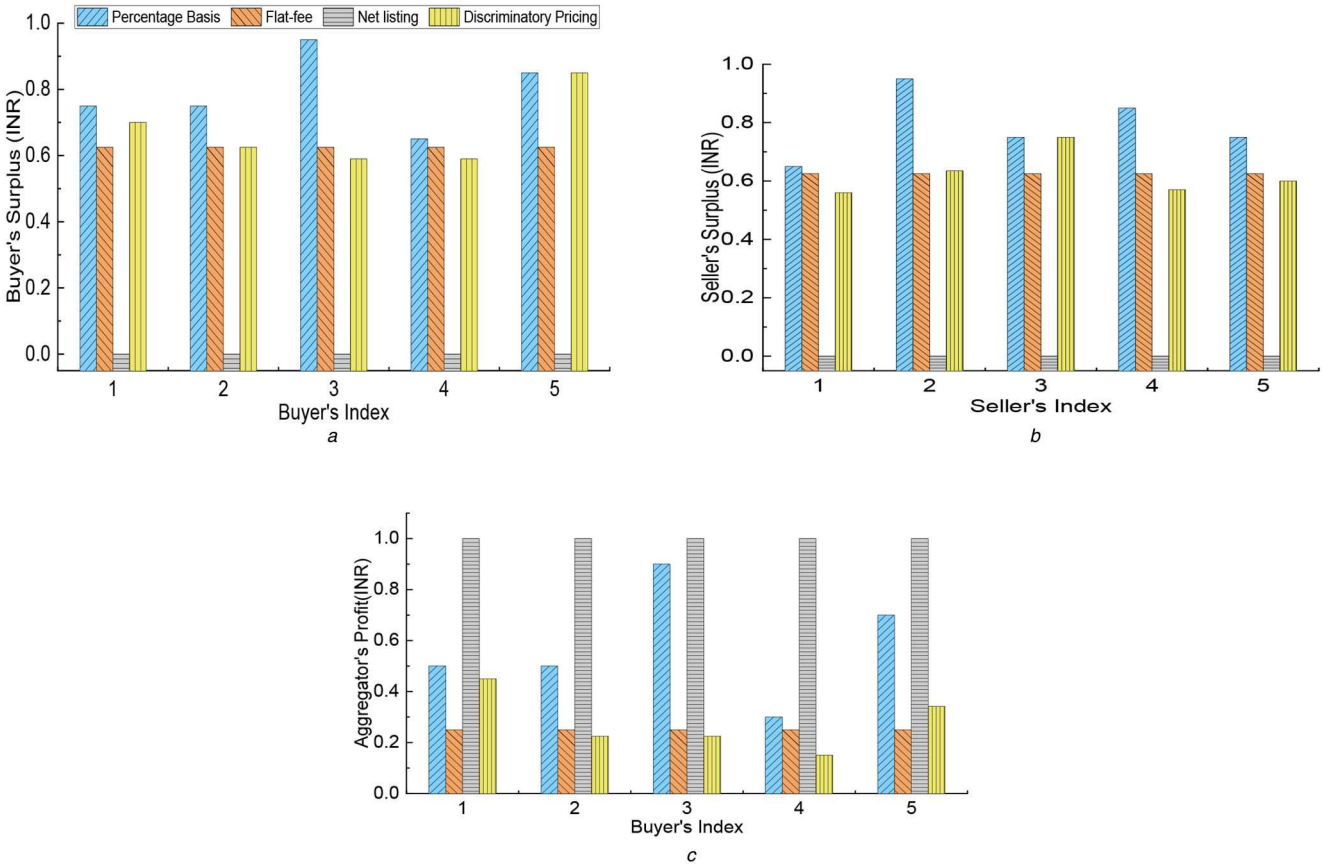

Fig. 10 *Comparison of buyer–seller surplus and aggregator*
(a) Buyer's surplus, ***(b)*** Seller's surplus, ***(c)*** Aggregator profit

Table 2 SSI and BSI under different pricing strategy

Matched result		Percentage basis		Flat fee		Net listing		Discriminatory pricing	
Buyer	Seller	BSI	SSI	BSI	SSI	BSI	SSI	BSI	SSI
B1	S3	1.33	1.37	1.26	1.31	1	1	1.30	1.37
B2	S5	1.33	1.37	1.26	1.31	1	1	1.26	1.3
B3	S2	1.23	1.23	1.14	1.15	1	1	1.13	1.15
B4	S1	1.48	1.65	1.45	1.62	1	1	1.41	1.56
B5	S4	1.26	1.28	1.18	1.20	1	1	1.26	1.19

but both strategies divide the surplus equally between buyers and sellers. However, it is found that aggregator profit is higher in percentage basis as compared to the flat fee, as shown in Fig. 10*c*. Discriminatory pricing gives preference to buyer–seller behaviour and sets unique prices. Therefore, buyer–seller surplus and aggregator profit vary more or less following the past characteristics of prosumers. However, the matching is skewed towards sellers as SSI > BSI, as the total number of buyers and sellers are equal, and preference was given randomly to sellers while matching.

In commission-based strategy, the flat fee is more prosumer centric as total surplus (20) is more than aggregator's profit, as shown in Fig. 11. It can also be observed that discriminatory pricing strategy considers the combined benefit of all commission-based strategy. It provides discriminatory prices to buyer–seller like net listing but with some surplus to buyer–seller like percentage base and flat fee. Therefore, discriminatory pricing offers the maximum benefit to buyer, seller and aggregator, as compared with the comission-based startegies in the present study.

5.6 Comparison with existing methods

Various auction-based methods are discussed in the literature [27, 28]. Continuous trading and double-sided auction are modelled to compare the auction-based model with developed sharing economy-based model. In continuous trading, buyers and sellers continuously place their orders and are matched on a continuous basis, such as stock exchanges. In the double-sided auction, trades are executed according to an order-driven system. Orders are matched for buyers and sellers for a particular time to maximise volume (Fig. 11).

Market clearing details for continuous trading are given in [28]. Modifications are done to remove network constraints from the algorithm. Input data of Table 1 is also modified as per assumed actual valuation of buyers and sellers. The bids are managed in descending order of the price, while asking price of sellers are arranged in ascending order for both the auctions as mentioned in Fig. 12. It can be seen that buyer B4 and seller S2 remain unmatched on both the auctions. To get matched, prosumers have to bid strategically in the auction like professional players. As compared to Algorithm 1 (Fig. 4), all the prosumers get matched and get benefited from LEM.

Table 3 provides the market clearing prices of the matched prosumers under (i) pay-as-bid pricing strategy and (ii) common clearing price (CCP). For pay-as-bid, auctioneer takes the payment difference between matched buyer–seller pairs. Prosumers do not get any surplus from LEM. CCP is calculated where buyer and seller prices intersect. Table 4 shows the SSI and BSI. In CCP, prosumers may get surplus if the difference between the bid/ask and CCP exists. In CCP strategy, auctioneer charges one-time platform joining fee from the prosumers which is usually high. Auction-based strategy is suitable to promote competition among

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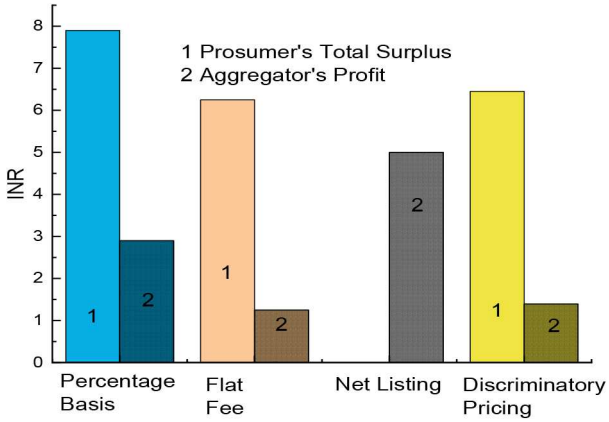

Fig. 11 *Prosumer's total surplus and aggregator's profit under different pricing strategies*

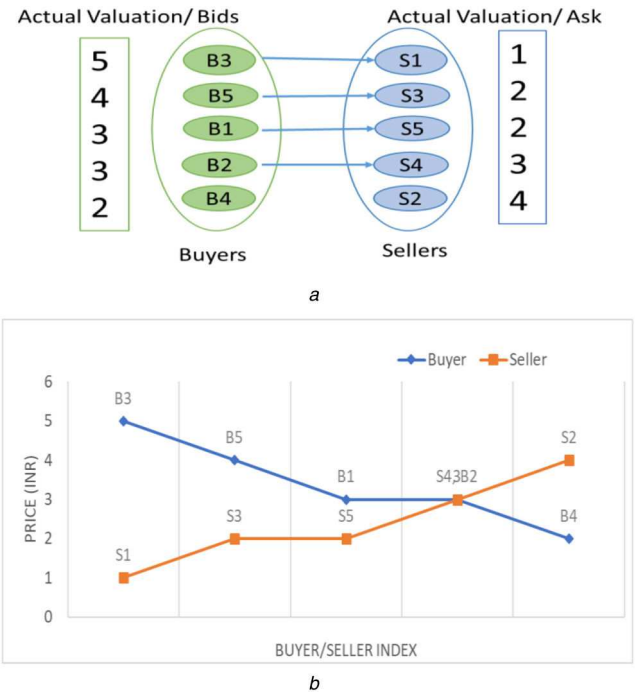

Fig. 12 *Buyer–seller matching in*
(a) Continuous trading, ***(b)*** Double-sided auction

Table 3 Clearing prices for continuous trading and double-sided auction

Matched result		Clearing price, INR			
Buyer	Seller	Pay-as-bid		CCP	
		Buyer	Seller	Buyer	Seller
B3	S1	5	1	3	3
B5	S3	4	2	3	3
B1	S5	3	2	3	3
B2	S4	3	3	3	3

Table 4 SSI and BSI under continuous trading and double-sided auction

Matched result		Continuous trading		Double-sided auction	
Buyer	Seller	BSI	SSI	BSI	SSI
B3	S1	1	1	1.66	3
B5	S3	1	1	1.33	1.5
B1	S5	1	1	1	1.5
B2	S4	1	1	1	1

prosumers. Moreover, prosumers are less motivated in continuous trading as compared with double-sided auction. Comparisons tabulated in Tables 2 and 4 present uneven distribution of surplus among matched, which hinders prosumer centricity.

6 Conclusion

Energy sharing in P2P LEM framework has the potential for prosumers integration on distribution network in future. This would bring significant economic benefit to prosumers as well as help to

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utilise energy resources locally. LEM organisation in the existing market is discussed with various upcoming market entities in the energy market. This paper proposes a P2P LEM platform, its market integration and different pricing strategies for aggregator's two-sided platform. Aggregator owns a two-sided market platform and acts as matchmaking intermediary for buyer and seller peers.

The general framework of matching and pricing provided in the paper, to devise P2P LEM platform, is easy to be adapted for initial prosumer integration in future power systems. Moreover, aggregator's profit is calculated under different pricing strategies. The choice of pricing-strategy depends on the number of buyer–seller pairs in the market and performance indices of prosumers, motivating them to connect with the platform. The proposed P2P platform is compared with the existing continuous trading platform. The key benefit of the proposed platform is that the maximal number of matching can be provided to satisfy prosumers. One can consider and model the effect of entry and exit time of prosumers while matching the prosumers, for future research.

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